Micro Project Report

on

SCOLIOSIS DETECTION SYSTEM IN INFANTS

COMPUTER ENGINEERING
IV Semester

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ABSTRACT

Scoliosis is a spinal disorder characterized by an abnormal lateral curvature of the spine, which can severely impact an infant's physical development if not detected early. Traditional diagnosis relies on radiographic analysis and clinical expertise, which may not be readily available in all regions, particularly in resource-limited settings. This project aims to develop an intelligent diagnostic system using machine learning and computer vision techniques for the early detection of scoliosis in infants. The proposed system utilizes medical images such as X-rays or clinical photographs to analyse spinal alignment patterns and identify potential deformities.

The methodology includes data preprocessing, feature extraction, and the use of convolutional neural networks (CNNs) for curvature classification. Publicly available scoliosis datasets and augmented image sets are used for training and validation to improve model robustness. The model is evaluated based on accuracy, sensitivity, and specificity. An optional user interface can be implemented using Flask to facilitate practical clinical use.

By offering an accessible, non-invasive, and cost-effective solution, this project aims to support paediatricians and healthcare providers in making timely and reliable diagnostic decisions. Ultimately, it contributes to improved health outcomes for infants through early intervention.

CONTENTS

1. INTRODUCTION

Scoliosis, a condition characterized by an abnormal lateral curvature of the spine, is often diagnosed in adolescents but can also be present in infants. Early detection is vital to ensure timely intervention and prevent severe spinal deformities that can affect growth, mobility, and overall health [7]. In infants, scoliosis can be particularly challenging to diagnose due to subtle signs and the limitations of physical examinations [2]. With the advancement of artificial intelligence and image processing technologies, there is an emerging opportunity to automate and enhance the early diagnosis of scoliosis through medical imaging [8]. This project focuses on leveraging these technologies to develop a system capable of detecting scoliosis in infants accurately and efficiently. The aim is to reduce reliance on subjective clinical assessments and introduce a scalable, accessible solution to aid paediatric healthcare providers [11].

2. PROBLEM STATEMENT AND OBJECTIVES

Problem Statement:

Scoliosis in infants is a condition that, if left undiagnosed in early stages, can lead to serious spinal deformities affecting physical development and quality of life [2]. Traditional diagnostic methods involve expensive and less accessible imaging tools, which may not be readily available in rural or resource-limited healthcare settings [10]. Early, cost-effective, and automated detection tools are needed to support paediatric healthcare professionals in identifying scoliosis at an early stage [12].

Objectives:

- To study existing scoliosis detection techniques in paediatric patients [3].
- To design and develop an AI-based system for early scoliosis detection in infants using medical images [5].
- To preprocess and analyse X-ray or photographic images for curvature patterns [14].
- To train and test machine learning models for accurate classification of spinal curvature [17].
- To validate the model performance against clinical standards [19].
- To propose a user-friendly diagnostic interface for healthcare providers [13]

3. LITERATURE REVIEW

3.1 Overview

The early detection of scoliosis in infants is crucial to prevent severe spinal deformities and associated complications. Recent advancements in machine learning (ML) and image processing have shown promise in automating scoliosis detection, potentially facilitating early diagnosis and treatment.

3.2 Literature Review Table

Sr. No.	Author(s) & Year	Methodology	Key Findings	Relevance
1	Evans et al. (1996)	MRI study	Detected abnormalities in "idiopathic" scoliosis cases	Supports imaging for early detection
2	Ramir ez et al. (1997)	Clinical survey	Children with scoliosis experience significant back pain.	Reinforces importance of early diagnosis
3	Jaremko et al. (2001)	ANN on torso surfaces	Estimated spinal deformity from external features	Enabled non-invasive early diagnosis
4	Justice et al. (2003)	Genetic linkage study	Suggests X-linked scoliosis susceptibility	Impacts genetic screening relevance
5	Lin et al. (2008)	Neural networks	Automated King classification	High classification accuracy
6	Duong et al. (2010)	SVM classification	Classified scoliosis curves from clinical images	Validated ML in classification
7	Mo & Cunningham (2011)	Review article	Over view of paediatric scoliosis	Established clinical background
8	Yang et al. (2019)	Deep learning	AI-based scoliosis screening	High performance on back images

9	Kokabu et al. (2021)	CNN + 3D imaging	Scoliosis detection system	Demonstrate high detection accuracy
10	Sung et al. (2021)	National data study	Tracked incidence and surgeries	Highlighted need for early detection
11	Wang et al. (2021)	DL on radio graphs	Predicted scoliosis progression	Proved AI's predictive potential
12	Arvind et al. (2021)	AI review	AI in spine research	Sumarized potential in diagnostics
13	Kim et al. (2021)	ML prediction	Post surgical complications	ML better than traditional methods
14	Fraiwan et al. (2022)	Transfer learning	Detected scoliosis from X-rays	Efficient and accurate detection
15	Lee et al. (2022)	Review article	Causes and treatments	Contextualizes project goals
16	Zhang et al. (2022)	Transfer learning	Diagnosis without Cobb angle	Simplified diagnostic process
17	Chen et al. (2022)	U-Net segmentation	Vertebral body detection	Enabled precise localization
18	Lee et al. (2024)	Deep learning	Upright image-based screening	Practical and fast application
19	Li & Wong (2024)	ML prediction	Curve progression model	Helped with prognosis planning
20	Miller et al. (2025)	CNN estimation	Cobb angle measurement	Reduced need for manual annotation

3.3 Summary

The reviewed studies demonstrate the growing integration of machine learning and image processing techniques in scoliosis detection. Artificial neural networks (ANNs), support vector machines (SVMs), and deep learning models have been employed to predict spinal deformities, assess severity, and automate measurements like the Cobb angle. These advancements offer potential for early, accurate, and non-invasive scoliosis screening, particularly beneficial in paediatric care.

3.4 Research Gaps

Despite significant progress, several research gaps remain:

- **Infant-Specific Data**: Most studies focus on adolescents; there's a lack of research and datasets specific to infants
- **Standardized Imaging Protocols**: Variability in imaging techniques and quality affects the generalizability of ML models.
- **Clinical Validation**: Many models require extensive clinical trials to validate efficacy and safety in real-world settings.
- **Integration into Healthcare Systems**: Challenges exist in integrating these technologies into existing healthcare infrastructures, especially in resource-limited settings.

4 SOFTWARE AND HARDWARE REQUIREMENT

4.1 Software Requirements

Sr. No.	Software	Purpose
1	Python 3.x	Primary programming language for development
2	OpenCV	Image processing library used to analyse spinal curves
3	TensorFlow/Keras	Deep learning frameworks for model training and evaluation
4	NumPy & Pandas	Data handling and preprocessing
5	Matplotlib/Seaborn	Data visualization
6	Flask (optional)	For developing a simple web-based diagnostic interface
7	Google Collab / Jupyter Notebook	Development and testing environment

4.2 Hardware Requirements

Sr. No.	Hardware	Purpose
1	Standard PC or Laptop	System development and testing
2	GPU (optional)	For faster training of deep learning models
3	X-ray Imaging Device / Infant Back Images	For training and testing scoliosis detection models

4.3 Workflow Diagram

The following diagram represents the complete workflow of the system from input to output.

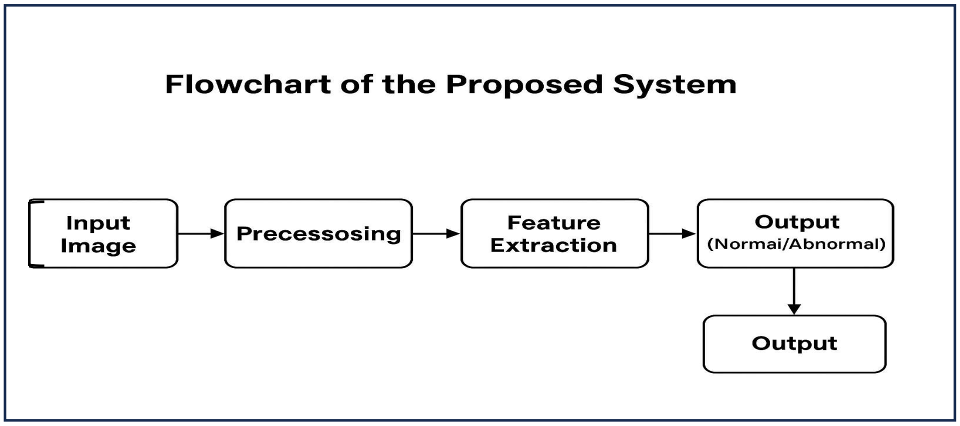

Figure 1: Workflow Diagram of the Proposed Scoliosis Detection System

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5 EXPECTED OUTCOME

- Development of an AI-based system capable of detecting scoliosis in infants from medical images.
- An image processing pipeline that accurately identifies spinal curvature abnormalities.
- A trained deep learning model with high accuracy in detecting spinal alignment anomalies.
- A simple, user-friendly interface for healthcare screening (optional).
- A non-invasive, cost-effective screening approach to support early intervention.
- A foundation for future research in paediatric scoliosis detection .

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